

Nutrition in Diabetes

Note: This reference document contains the content of the Canvas page for “Nutrition in Diabetes” with minor modifications for presentation in a document format. Refer to this session’s Canvas page for the full interactive experience or if any hyperlinks do not function properly.

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Overview

One hallmark of diabetes mellitus is elevated glucose in blood, even though the underlying cause differs between the different types of diabetes. A major goal of nutritional and other management is to ensure that glucose levels stay within reasonable levels. Additional goals are to get close to normal weight (BMI) in patients. The material presented here will discuss principles and practical approaches to achieve these goals.

Learning Objectives

By the end of this session, you will be able to meet the following learning objectives:

1. Identify and compare the treatment goals for obese, type 2 and insulin-dependent diabetics.
2. Discuss signs, symptoms and management of hypoglycemia in diabetes.
3. Describe the practical approaches in the management of diabetes through diet and exercise.
4. Discuss the changes in insulin with food of varied glycemic index and compare it to glycemic load of those foods.
5. Explain the effects of alcohol intake in diabetes.
6. Discuss the team approach to patient care identifying the roles of the physician, nurse, dietitian, the patient educator and exercise therapist.
7. Explain the role of exercise in controlling blood glucose levels, and in the prevention of some of the complications of diabetes.
8. Discuss gestational diabetes and management with diet and exercise.
9. Explain the reason for appropriate preventative action during illness or infection in the management of diabetes.

Prerequisites

Before progressing with this session, please review the following materials:

Semester 4, GINU

- Enhancement Session | Hormonal Regulation of Carbohydrate Metabolism 1
- Enhancement Session | Hormonal Regulation of Carbohydrate Metabolism 2

Please also be reminded of these laboratory values which you need to be able to apply:



Glucose	
Fasting	70–100 mg/dL
Random, non-fasting	<140 mg/dL

Hemoglobin A1c (HbA1c)	
Normal	<5.6%
Prediabetes	5.7% - 6.4%
Diabetes	>6.5%

Please also be reminded that a treatment goal for a patient can exceed the values found in non-patients.

Resources

Required

Textbook: [Essentials of Clinical Nutrition in Healthcare \(7th ed.\)](#)

- Chapter 18. Testable material is the entire chapter except the section on Basic Epidemiology.

Optional

- For those wanting more information, UpToDate has an expanded discussion of this topic. However, the chapter listed above is sufficient for the testable material.